

Xinran Zhou

zhou.xinran@colostate.edu | (970) 825-9190 | Fort Collins, CO 80521

PROFESSIONAL SUMMARY

Graduate student in carbon management and environmental economics with experience in greenhouse gas accounting, life cycle assessment, and environmental-economic analysis. Uses IPCC methods, quantitative modeling, and reproducible data workflows to support low-carbon decision-making in agriculture and energy systems. Strong background in cross-disciplinary research, technical writing, and bilingual communication.

TECHNICAL SKILLS

- **Environmental Assessment:** Life Cycle Assessment (LCA), Life Cycle Costing (LCC), GHG inventories, Input-Output Analysis
- **Programming & Data Analysis:** R, MATLAB, Stata
- **LCA & Sustainability Software:** SimaPro, OpenLCA, COMET-Farm
- **Data Visualization & Reporting:** Microsoft Excel, Origin, reproducible reporting (R Markdown)
- **Languages:** Mandarin Chinese (native), English (fluent)

EDUCATION

Colorado State University, Fort Collins, CO	Jan 2026 – Jan 2027 (expected)
<i>Professional Science Master, Ecosystem Science & Sustainability (Carbon Management)</i>	
Shandong University, Weihai, China	Sep 2024 – Jun 2027 (expected)
<i>M.S., Population, Resources & Environmental Economics GPA: 93.55/100</i>	
China University of Geosciences (Wuhan), Wuhan, China	Sep 2020 – Jun 2024
<i>B.A., Economics, with Distinction GPA: 4.05/5.0</i>	

RESEARCH EXPERIENCE

Course Project

Environmental Data Science Applications Fort Collins, CO	Jan 2026 – Present
• Clean, analyze, and visualize environmental and economic datasets in R to support data-driven sustainability assessments. • Identify trends in emissions, environmental indicators, and sustainability metrics using statistical methods. • Produce reproducible reports and data visualizations to clearly communicate key analytical findings.	

Greenhouse Gas Inventory Analysis Fort Collins, CO	Jan 2026 – Present
• Estimate GHG emissions using IPCC-based methods and regression- or process-based models to support evidence-based mitigation planning. • Compile inventories and conduct mitigation analyses in R and COMET-Farm, including scenario design and marginal abatement cost estimation. • Develop mitigation strategies by assessing technical potential and socioeconomic constraints across alternative scenarios.	

Research Assistant

LCA-LCC Assessment of SHS Production and Agricultural Application China	Oct 2024 – Present
• Conduct a first-author LCA-LCC study of four hydrothermal humification technologies for Synthetic Humic Substance (SHS) production in China. • Quantify life-cycle environmental impacts, economic costs, and process/material hotspots to identify high-impact improvement opportunities. • Evaluate province-level application benefits and future mitigation potential under regional energy transition scenarios to inform low-carbon deployment pathways.	
Construction of Chinese Food and Agriculture Biomass Input-Output (FABIO) Model China	Oct 2024 – Oct 2025
• Collected and cleaned multi-source agriculture and biomass datasets to support model construction and database consistency. • Built and documented an environmental satellite account by mapping sectoral environmental extensions to 84 detailed agricultural products.	

- Performed quality control and consistency checks on units, regional and sector mapping, and time-series comparability to improve data reliability and model transparency.

Research on a Hubei-Specific ESG Evaluation and Application System | China

Sep 2022 – Jun 2024

- Collected and analyzed data for ESG evaluation research to support institutional reporting and framework development.
- Contributed to the ESG report for China University of Geosciences (Wuhan).
- Supported the development of a Hubei-specific ESG framework based on global evaluation systems.

Research on Green Energy Policy Effects on Wind Power Development | China

Sep 2022 – Sep 2023

- Reviewed literature, collected data, and developed analytical models to evaluate how green energy policies influence wind power development.
- Contributed to a paper on the impact of industrial policies on the sustainable development of wind power enterprises and the underlying mechanisms involved.
- Won third prize in the National Energy Economics Innovation Competition (Provincial-level).

Research on Pricing Issues in China's Downstream Natural Gas Market | China

Sep 2021 – Sep 2022

- Built **NetLogo** simulations to analyze how natural gas prices affect user energy conservation behavior and downstream market responses.
- Conducted customer value analysis, managed research data, and contributed to a paper using Gower's similarity coefficients and the PAM clustering algorithm.
- Won second prize in the National Energy Economics Innovation Competition (National-level).

Project Leader

Research on Multi-Actor Strategic Interactions in Green Energy Policy | China

Jun 2023 – Jun 2024

- Led a national-level student research project and built evolutionary game models in **MATLAB** to analyze interactions among government, green power producers, and consumers under Renewable Portfolio Standards.
- Examined multi-actor interest negotiation and policy responses in green energy policy design through model-based analysis.
- Authored a first-author paper on multi-actor strategic interactions under Renewable Portfolio Standards.

ACTIVITIES

Technical Translation & Localization Intern

Jul 2025 – Aug 2025

Beijing Peiqi Technology Co., Ltd. | China

- Completed training in technical documentation and localization workflows.
- Translated and proofread technical materials between Chinese and English with consistent terminology and clear wording.
- Organized internal documentation and coordinated revisions with team members to meet deadlines.

Part-time Bilingual Economics Instructor (A-Level Economics)

Nov 2023 – Dec 2024

Suzhou Haitong Education & Culture Consulting Co., Ltd. | China

- Taught A-Level Economics in Chinese and English, covering both microeconomics and macroeconomics.
- Developed lesson plans and exam-focused practice materials aligned with course requirements.
- Provided individualized feedback and tracked student progress.

Communications & Media Coordinator

Sep 2020 – Jun 2021

Publicity Team, China University of Geosciences (Wuhan) | China

- Wrote and edited event news briefs and announcements for publication.
- Formatted content to maintain consistency and publication quality.
- Scheduled and published posts on the official WeChat account while coordinating materials and deadlines with the team.

HONORS & COMPETITIONS

- **National Scholarship** (offered by Ministry of Education of the People's Republic of China, 2023)
- National Energy Economics Innovation Competition, Second Prize (National-level, 2022)
- International College Students' "Internet+" Innovation & Entrepreneurship Competition, Bronze Award (National-level, 2021)
- National English Debate Competition for College Students, Third Prize (National-level, 2021)
- Outstanding Graduate (2024); Outstanding Student Leader (University-level, 2021 & 2022); Outstanding Volunteer (University-level, 2022); Outstanding Intern (College-level, 2021)